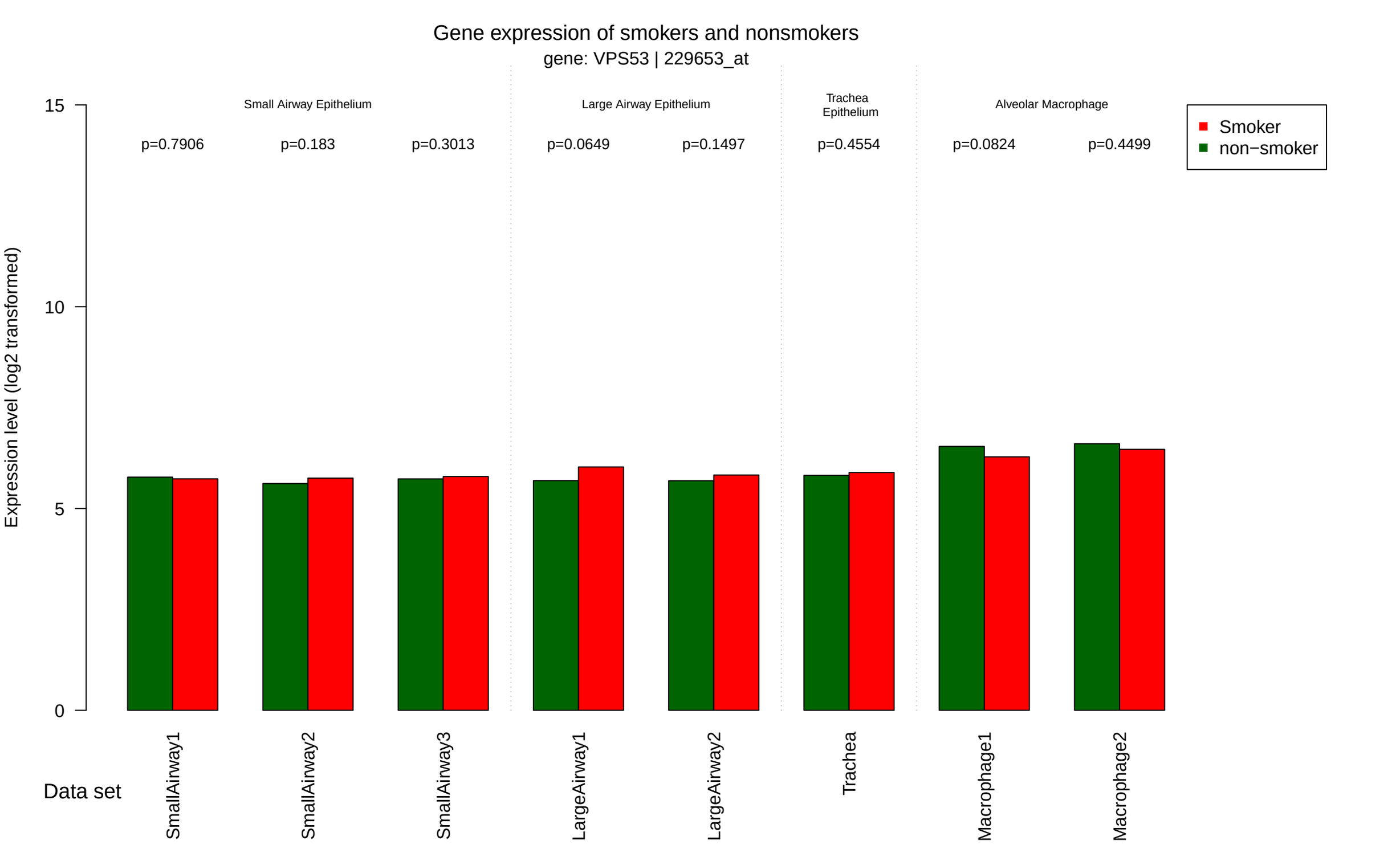
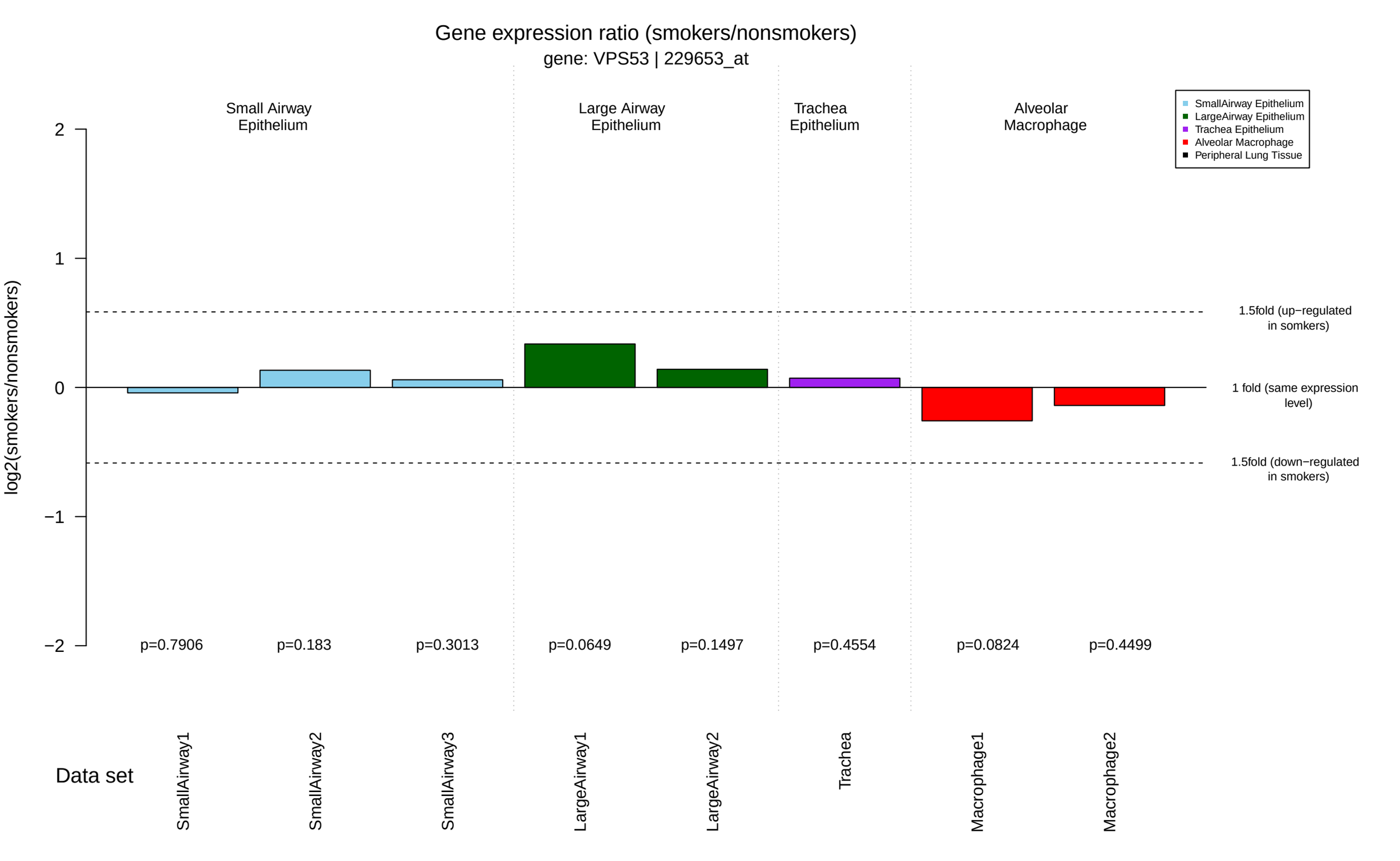
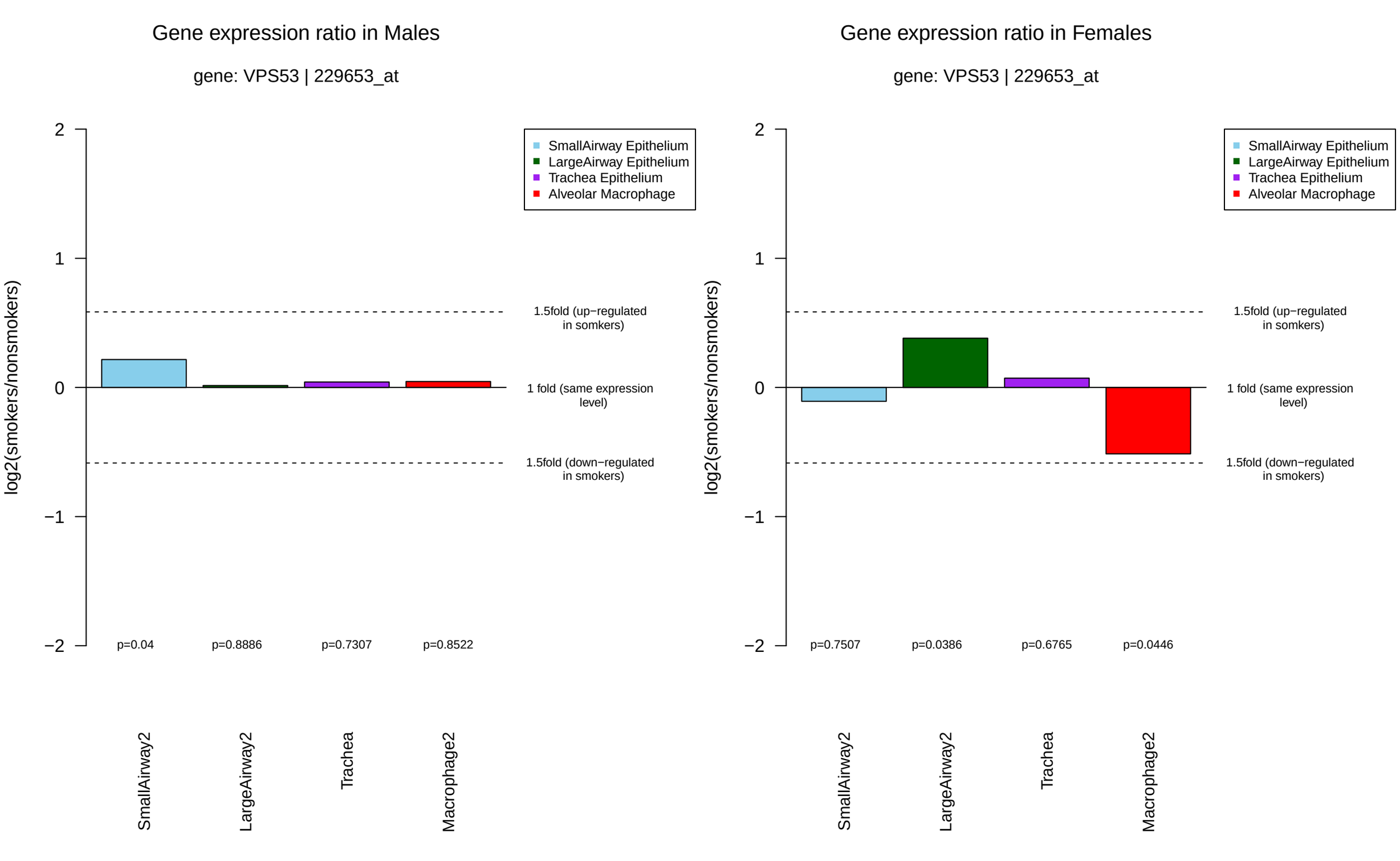
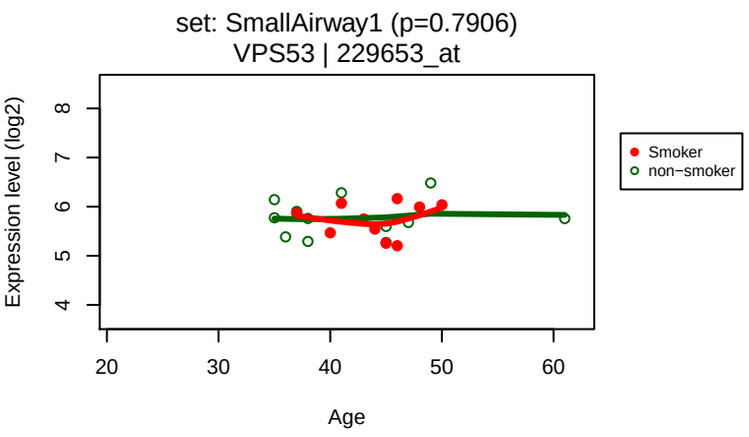
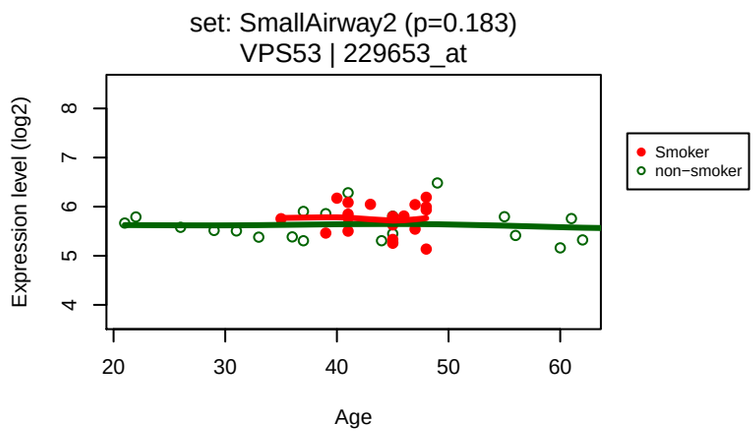
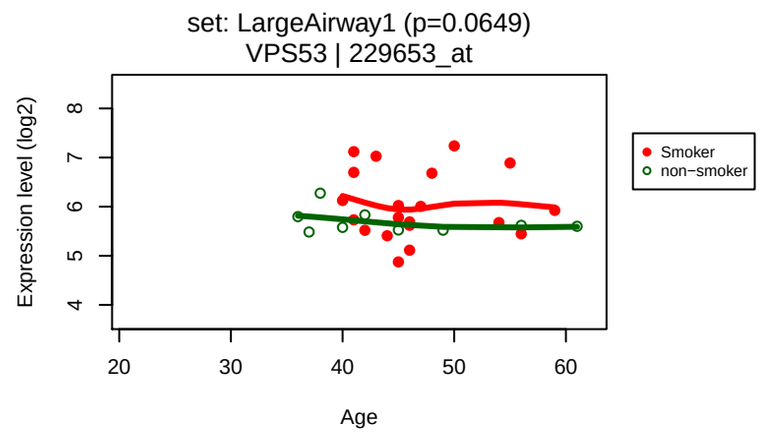

Gene expression of smokers and nonsmokers
gene: VPS53 | 229653_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7906
p=0.183
p=0.3013
p=0.0649
p=0.1497
p=0.4554
p=0.0824
p=0.4499
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

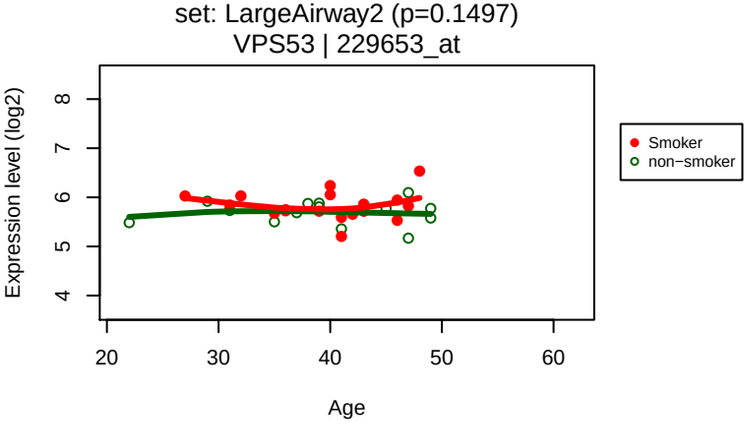

Gene expression ratio (smokers/nonsmokers)
gene: VPS53 | 229653_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7906
p=0.183
p=0.3013
p=0.0649
p=0.1497
p=0.4554
p=0.0824
p=0.4499
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

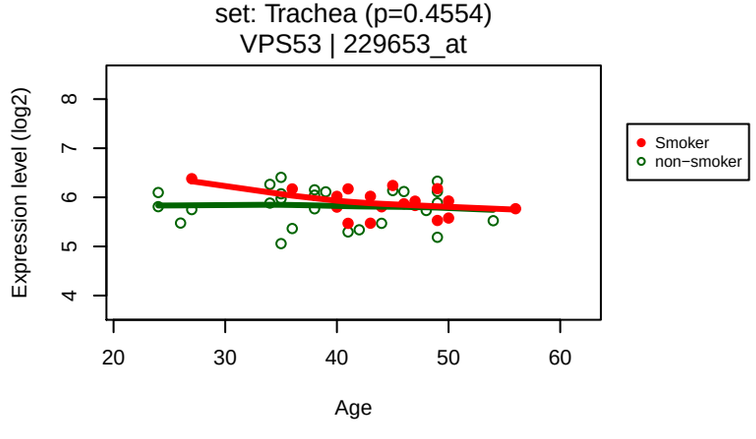

Gene expression ratio in Males
gene: VPS53 | 229653_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.04
p=0.8886
p=0.7307
p=0.8522
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: VPS53 | 229653_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7507
p=0.0386
p=0.6765
p=0.0446
SmallAirway2
LargeAirway2
Trachea
Macrophage2

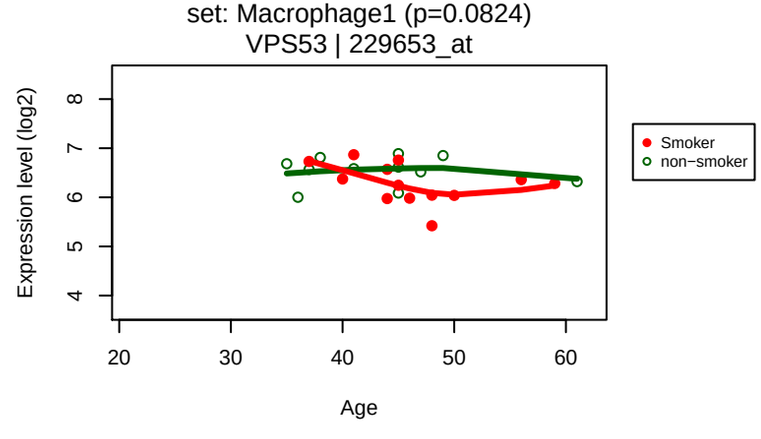

set: SmallAirway1 (p=0.7906)
VPS53 | 229653_at
Expression level (log2)
Age
Smoker
non−smoker

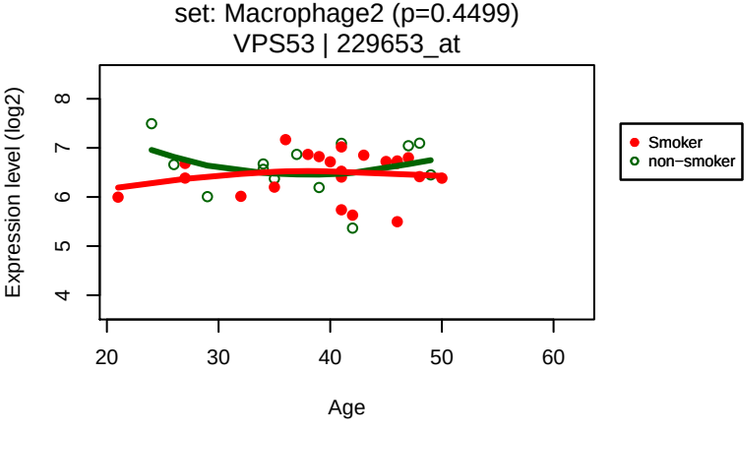

set: SmallAirway2 (p=0.183)
VPS53 | 229653_at
Expression level (log2)
Age
Smoker
non−smoker

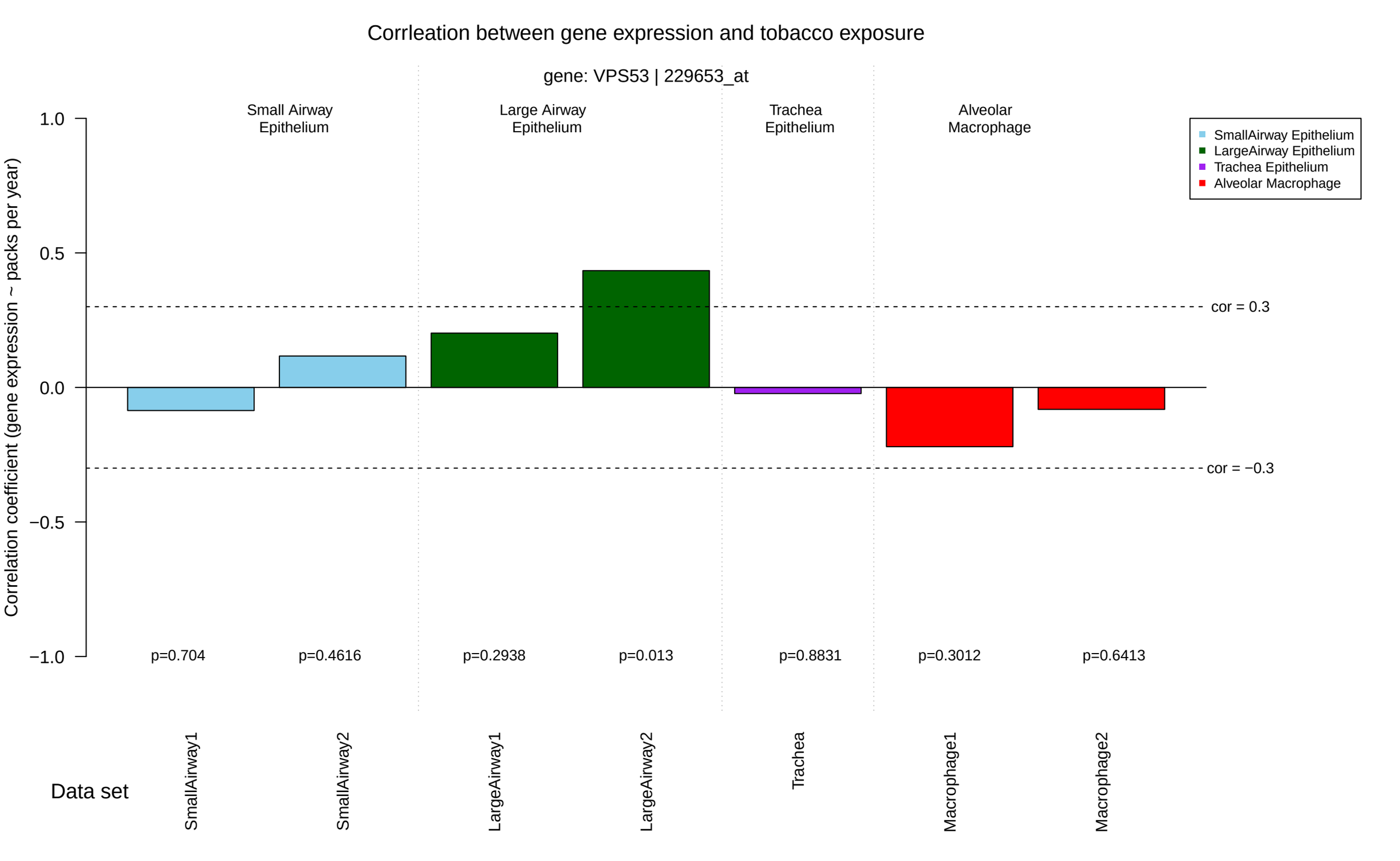

set: LargeAirway1 (p=0.0649)
VPS53 | 229653_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1497)
VPS53 | 229653_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4554)
VPS53 | 229653_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0824)
VPS53 | 229653_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4499)
VPS53 | 229653_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: VPS53 | 229653_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.704
p=0.4616
p=0.2938
p=0.013
p=0.8831
p=0.3012
p=0.6413
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2